Lab 5: Static Code Analysis

Fitness Club Class Booking Ledger

Python File: gym_system.py

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1. About the Program

The Fitness Club Class Booking Ledger is a Python program that manages bookings for fitness classes such as Yoga, HIIT, and Spin. It supports member bookings, cancellations, class capacity management, guest bookings, displaying class rosters, and exporting booking information to a JSON file. The original program contained code quality, logic, security, and style issues. This lab uses Pylint, Bandit, and Flake8 to identify and fix these issues.

2. Objective

To improve Python code quality, security, style, and maintainability using static code analysis tools.

3. Tools Used

- Pylint: Identifies potential programming issues and code quality problems.
- Bandit: Detects potential security vulnerabilities in Python code.
- Flake8: Checks Python formatting and compliance with style guidelines.

4. Procedure

1. Inspected gym_system.py and ran the original program.
2. Installed Pylint, Bandit, and Flake8 in the Python virtual environment.
3. Ran all three tools on the original code and saved the reports.
4. Examined the reports and documented the issues.
5. Fixed the identified problems in gym_system.py.
6. Reran the tools to verify the improvements.

5. Issues Identified and Fixed

The table below documents the issues found in the original code, their types, original line numbers, descriptions, and fix approaches.

Issue	Type	Line(s)	Description	Fix Approach
Shell command execution	Security — High	36	os.system() executes a constructed shell command, creating a potential command-injection vulnerability.	Replaced os.system() with safer file operations using pathlib.
Mutable default argument	Bug	8	attendance_log=[] shares a default list across function calls.	Changed the default to None and initialized a new list inside the function.

Incorrect capacity check	Logic bug	12	Using > instead of >= allows an extra booking when the class reaches its capacity.	Changed the condition to >= to prevent overbooking.
Bare exception	Code quality	21	A bare except can catch unrelated exceptions and hide errors.	Replaced it with except (KeyError, ValueError).
Missing file encoding	Code quality	32	The file was opened without explicitly specifying an encoding.	Added encoding='utf-8'.
File resource handling	Code quality	32	The file was opened without a context manager.	Used with open(...) for proper resource management.
Invalid class capacity	Input validation	Multiple	Negative class capacities were accepted.	Added validation for non-negative integer capacities.
Invalid guest count	Input validation	Multiple	Invalid guest counts could cause unexpected behavior.	Added validation for non-negative integer counts.
Missing docstrings	Maintainability	Multiple	The module and functions lacked documentation.	Added module and function docstrings.
Naming convention	Style	27	addMembers did not follow Python's snake_case naming convention.	Renamed it to add_members.
Formatting and dictionary iteration	Style	15, 39	Old string formatting and dictionary key iteration reduced readability.	Used f-strings and .items() where appropriate.

Note: Line numbers refer to the original code and may differ from the corrected version. Verify individual line numbers against the original reports before submission.

6. Static Analysis Results

Tool	Before Fixes	After Fixes
Pylint	6.60/10	10.00/10
Bandit	1 high-severity finding	No issues identified
Flake8	10 reported violations	No violations reported

7. How to Run the Program

Prerequisites

Python installed on your system and a project virtual environment.

Execution Steps

7. Open a terminal in the project directory.
8. Activate the virtual environment: source venv/bin/activate
9. Run the program: python gym_system.py
10. Enter a non-negative whole number for walk-in guests, or press Enter to use the default value of 1.

The program displays class rosters and exports them to gym_schedule.json.

Run Static Analysis

Run these commands to generate the final reports:

```
pylint gym_system.py > pylint_report_after.txt 2>&1

bandit -r gym_system.py > bandit_report_after.txt 2>&1

flake8 gym_system.py > flake8_report_after.txt 2>&1
```

8. Project Files

- gym_system.py — Corrected Python application.
- issues.md — Detailed issue log and fixes.
- pylint_report.txt — Original Pylint report.
- pylint_report_after.txt — Final Pylint report.
- bandit_report.txt — Original Bandit report.
- bandit_report_after.txt — Final Bandit report.
- flake8_report.txt — Original Flake8 report.
- flake8_report_after.txt — Final Flake8 report.
- reflection.md — Reflection answers required by the lab.

9. Conclusion

This lab demonstrated how static code analysis can identify and help resolve programming, security, input validation, and style issues. After the improvements, the program achieved a Pylint score of 10.00/10, Bandit reported no issues, and Flake8 reported no violations. These improvements make the Fitness Club Class Booking Ledger easier to read, maintain, and review for potential security issues.